Project 2: Parameter Estimation

Math 540: Uncertainty Quantification

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Problem 1: Heat Model Parameter Estimation (Copper Rod)

Problem Statement

Repeat the analysis of Example 11.22 for the steady state heat model using the copper data in Table 1 and the values $k = 4.01$ for the thermal conductivity and $u_{amb} = 22.28$. Do your residuals appear to be iid? Use a Q-Q plot to establish whether the residuals are normally distributed.

x	10	14	18	22	26	30	34	38	42	46	50	54	58	62	66
Temp	66.04	60.04	54.81	50.42	46.74	43.66	40.76	38.49	36.42	34.77	33.18	32.36	31.56	30.91	30.56

Table 1: Steady-state temperatures measured at locations x for a copper rod.

Solution

We are given the temperature of the copper rod at equidistant points. Using the analytic solution of the heat equation and given data, we constructed a least squares problem. Using `fminsearch.m`, we obtained parameter estimates:

$$\Phi = -9.9265, \quad h = 0.0014$$

Model Fitting

The graph of our model using estimated parameters and data shows good fitting. However, we observe from the residual plot that residuals are identical but not independent. The Q-Q plot analysis shows that residuals do not follow a straight line, so residuals are not from a normal distribution.

Variance and Covariance Estimation

With $n = 16$ data points and $p = 2$ parameters, we compute the error variance estimate:

$$\sigma^2 = \frac{1}{n-p}\mathbf{R}^\top\mathbf{R} = 0.0529$$

where $\mathbf{R}$ is the vector of residuals.

Using the sensitivity matrix X, we obtain the covariance matrix:

$$V = \sigma^2[X^\top X]^{-1} = \begin{bmatrix} 0.0090 & -1.2117 \times 10^{-6} \\ -1.2117 \times 10^{-6} & 1.7726 \times 10^{-10} \end{bmatrix}$$

Sensitivity Phi

Figure 1: Sensitivity Phi

Sensitivity h

Figure 2: Sensitivity h

Standard Deviations

The estimated standard deviations are:

$$\sigma = 0.2301, \quad \sigma_\Phi = 0.0947, \quad \sigma_h = 1.3314 \times 10^{-5}$$

95% Confidence Intervals

Using the t-distribution:

$$\Phi \in [-10.1310, -9.7219]$$

$$h \in [0.0014, 0.0015]$$

Figure 1: Sensitivity equations: (a) $\frac{\partial u}{\partial \Phi}$; (b) $\frac{\partial u}{\partial h}$.

Figure 2: (a) The graph of the data and model for the uninsulated copper rod; (b) The residual plot; (c) Q-Q plot of the residuals.

Problem 2: Helmholtz Energy Model (OLS Estimation)

Problem Statement

Consider the Helmholtz energy:

$$\psi(P, \theta) = \alpha_1 P^2 + \alpha_{11} P^4 + \alpha_{111} P^6$$

where P is the polarization on the interval $[0, 0.8]$ and $\theta = [\alpha_1, \alpha_{11}, \alpha_{111}]^\top$ are parameters with nominal values $\alpha_1 = -389.4$, $\alpha_{11} = 761.3$, $\alpha_{111} = 61.5$.

Part (a): Variance Estimation for Different Sample Sizes

For $n = 81, 161, 801$ equally spaced polarization values $P_i = (i-1)\Delta P$, $\Delta P = \frac{0.8}{n-1}$, we compute the model response and observations:

Model Fit

Figure 3: Model Fit

Residuals

Figure 4: Residuals

Q-Q Plot

Figure 5: Q-Q Plot

$$Y_i = \psi(P_i, \theta) + \varepsilon_i$$

where $\varepsilon_i \sim_{iid} N(0, \sigma^2)$ with $\sigma = 2.2$.

The OLS estimate for observation variance is:

$$\hat{\sigma}^2 = \frac{1}{n-p}\mathbf{R}^\top \mathbf{R}$$

Results:

n	$\hat{\sigma}$
81	2.3841
161	2.2594
801	2.1791

As data points increase, the estimated variance converges to the true variance $\sigma = 2.2$.

Part (b): Normal Equations and Parameter Estimation

Using the observation model:

$$Y_i = \psi(P_i, \theta) + \varepsilon_i = [P_i^2, P_i^4, P_i^6] \begin{bmatrix} \alpha_1 \\ \alpha_{11} \\ \alpha_{111} \end{bmatrix} + \varepsilon_i$$

The design matrix is:

$$X = \begin{bmatrix} P_1^2 & P_1^4 & P_1^6 \\ P_2^2 & P_2^4 & P_2^6 \\ \vdots & \vdots & \vdots \\ P_n^2 & P_n^4 & P_n^6 \end{bmatrix}$$

The normal equations give the estimated parameters:

$$(X^\top X)\hat{\theta} = X^\top Y \implies \hat{\theta} = (X^\top X)^{-1} X^\top Y$$

Residual Plot

Figure 6: Residual Plot

Model and Observations

Figure 7: Model and Observations

$$\hat{\theta} = \begin{bmatrix} \hat{\alpha}_1 \\ \hat{\alpha}_{11} \\ \hat{\alpha}_{111} \end{bmatrix} = \begin{bmatrix} -390.0709 \\ 764.5939 \\ 57.2555 \end{bmatrix}$$

Covariance Matrix

$$V = \begin{bmatrix} 22.5543 & -110.0835 & 123.4682 \\ -110.0835 & 585.0489 & -690.2614 \\ 123.4682 & -690.2614 & 842.2265 \end{bmatrix}$$

Correlation Coefficients

The correlation coefficient between two random variables X and Y is:

$$\rho_{X,Y} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Computing correlations:

$$\rho_{\alpha_1,\alpha_{11}} = \frac{-110.0835}{\sqrt{22.543 \times 585.0489}} = -0.9585$$

$$\rho_{\alpha_1,\alpha_{111}} = \frac{123.4681}{\sqrt{22.543 \times 842.2265}} = 0.8960$$

$$\rho_{\alpha_{11},\alpha_{111}} = \frac{-690.2614}{\sqrt{585.0489 \times 842.2265}} = -0.9833$$

All parameters are mutually correlated with strong correlation. Thus, we cannot assume these parameters are mutually independent when employing global sensitivity analysis.

Figure 3: (a) Residual plot with 2σ intervals; (b) Graph of the model and observations.

Problem 3: SIR Model Parameter Estimation

Problem Statement

Consider the SIR model:

$$\frac{dS}{dt} = \delta N - \delta S - \gamma IS, \quad S(0) = 900$$

$$\frac{dI}{dt} = \gamma IS - (r+\delta)I, \quad I(0) = 100$$

$$\frac{dR}{dt} = rI - \delta R, \quad R(0) = 0$$

where $\gamma, r, \delta \in [0, 1]$.

Part (a): Parameter Estimation

Using `fminsearch.m` with the data from SIR.txt, we obtained:

$$\gamma = 0.0100, \quad \delta = 0.1953, \quad r = 0.7970$$

The residual plot shows that errors are identically and independently distributed. The fitted model shows good agreement with observational data.

Part (b): Sensitivity Analysis and Covariance

Using the complex-step derivative approximation:

$$f'(x) \approx \frac{\mathrm{Im}(f(x+ih))}{h}$$

with $h = 10^{-16}$, we obtained the sensitivity matrix χ.

With $n = 51$ data points and $p = 3$ parameters:

$$\sigma^2 = \frac{1}{n-p}\mathbf{R}^\top\mathbf{R} = 426.7780$$

Covariance Matrix:

$$V = \sigma^2(\chi^\top\chi)^{-1} = \begin{bmatrix} 1.7357\times10^{-8} & 1.8575\times10^{-7} & 1.4540\times10^{-7} \\ 1.8575\times10^{-7} & 2.6153\times10^{-5} & 3.1790\times10^{-5} \\ 1.4540\times10^{-7} & 3.1790\times10^{-5} & 8.4797\times10^{-5} \end{bmatrix}$$

Correlation Coefficients:

$$\rho_{\gamma,\delta} = 0.2757, \quad \rho_{\gamma,r} = 0.1198, \quad \rho_{\delta,r} = 0.6751$$

The parameters δ and r are correlated, but others are weakly correlated. The covariance matrix has rank 3, which is expected from Project 1.

Part (c): Parameter Distributions

Standard Deviations:

$$\sigma_\gamma = 1.3174 \times 10^{-4}, \quad \sigma_\delta = 0.0051, \quad \sigma_r = 0.0092$$

95% Confidence Intervals:

$$\gamma \in [0.0097, 0.0103]$$

$$\delta \in [0.1850, 0.2056]$$

$$r \in [0.7785, 0.8155]$$

Part (d): Influenza Outbreak Analysis

For the British boarding school influenza data with $N = 763$, $S(0) = 760$, $I(0) = 3$, $R(0) = 0$, and $\delta = 0$:

Day	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Confined	3	8	26	76	225	298	258	233	189	128	68	29	14	4

Table 2: Influenza data

Estimated Parameters:

$$\gamma = 0.0022, \quad r = 0.4469$$

With $n = 14$ points and $p = 2$ parameters:

$$\sigma^2 = 325.4893$$

Covariance Matrix:

$$V = \begin{bmatrix} 3.3117 \times 10^{-10} & 1.3030 \times 10^{-7} \\ 1.3030 \times 10^{-7} & 1.5355 \times 10^{-4} \end{bmatrix}$$

Correlation: $\rho_{\gamma,r} = 0.5778$

Standard Deviations:

$$\sigma = 18.0413, \quad \sigma_\gamma = 1.8198 \times 10^{-5}, \quad \sigma_r = 0.0124$$

95% Confidence Intervals:

$$\gamma \in [0.00217, 0.00226], \quad r \in [0.4215, 0.4755]$$

Code Files

File	Description
Problem1.m / Problem1.py	Heat model parameter estimation
Problem2.m / Problem2.py	Helmholtz energy OLS estimation
Problem3.m / Problem3.py	SIR model parameter estimation

References

1. Smith, R.C. (2013). *Uncertainty Quantification: Theory, Implementation, and Applications.* SIAM.
2. Banks, H.T. and Tran, H.T. (2009). *Mathematical and Experimental Modeling of Physical and Biological Processes.* CRC Press.